

SHM 377 Course Blueprint

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Course Purpose

SHM 377 develops the scientific, quantitative, regulatory, and professional reasoning needed to manage hazardous materials in workplace settings.

The course examines the management of hazardous materials across their lifecycle, including transportation, storage, use, exposure, release, emergency response, and disposal.

Students learn to characterize hazardous materials, understand how their physical and chemical properties influence their behavior, evaluate measurement and monitoring data, locate and apply regulatory and technical resources, and develop practical strategies for managing hazardous-material risks.

Rather than treating chemistry, regulations, exposure assessment, and data analysis as separate topics, SHM 377 integrates these tools through workplace problems and case studies.

By the end of the course, students should be able to recognize a hazardous materials problem, determine what information is needed, analyze relevant evidence and data, locate and apply appropriate technical and regulatory resources, and communicate a defensible management decision.

Catalog Description

Management of hazardous materials, including best practices and federal regulations associated with transportation, storage, use, and disposal of hazardous materials in the workplace.

Credits: 4

Contact Hours: 3 lecture, 2 laboratory

Prerequisites: CHEM 101 or CHEM 111/111LAB or CHEM 181/181LAB

Course Learning Outcomes

Students completing SHM 377 should be able to:

1. Anticipate, recognize, evaluate, and develop control strategies associated with hazardous materials.
2. Utilize federal regulations related to hazardous materials to assess compliance.
3. Utilize Department of Transportation regulations related to hazardous materials to assess compliance.

Fall 2026 Course Offering

SHM 377 is offered during Fall Quarter 2026 in Ellensburg.

Scheduled Meetings

Monday

- **Time:** 1:00 PM–3:20 PM
- **Location:** HOGU 226

Wednesday

- **Time:** 1:00 PM–3:20 PM
- **Location:** HOGU 226

The course will use these scheduled meetings to establish a recurring lecture–laboratory rhythm:

Monday
Concepts / Case Framing / Preparation
↓
Wednesday
Laboratory / Data / Application / Decision-Making

Where possible, Monday meetings should introduce the concepts, case context, technical tools, quantitative methods, or regulatory resources needed for the associated Wednesday laboratory or application.

Wednesday meetings should provide opportunities for students to apply those concepts through hazardous-material scenarios, quantitative analysis, measurement data, regulatory research, interpretation, and management decisions.

The Monday and Wednesday components should therefore function as an integrated instructional sequence rather than as separate courses.

Not every Wednesday meeting needs to produce a graded laboratory report.

Many class meetings may involve formative exercises, guided practice, calculations, regulatory navigation, data exploration, case studies, or project development without creating an additional graded assignment.

Enrollment

Fall 2026 enrollment should be monitored as registration changes.

The university course schedule currently lists a maximum enrollment of 36 students for SHM 377.

Course Philosophy

Hazardous-material management requires more than identifying a chemical or memorizing a regulation.

Professionals must be able to determine what material is present, understand how it behaves, determine where it may move, evaluate how it can be measured, interpret available data, identify applicable requirements, and decide how the material should be managed.

SHM 377 therefore uses an applied, case-based approach that integrates chemistry, exposure assessment, data analysis, regulations, and management decisions.

The course repeatedly asks students to move through a common reasoning process:

```
Material
  ↓
Phase / Form
  ↓
Medium
  ↓
Movement
  ↓
Properties
  ↓
```

Measurement

↓

Data

↓

Decision

The goal is not simply to memorize chemical properties, formulas, regulatory citations, labels, or tables.

The goal is to develop the ability to determine:

What is the problem?

↓

What information do I need?

↓

Where can I find that information?

↓

What does the evidence tell me?

↓

What requirements apply?

↓

What should we do?

↓

How can I defend and communicate that decision?

Hazardous Materials Reasoning Framework

Throughout the course, hazardous-material problems will repeatedly be examined using the following framework:

Material → Phase → Medium → Movement → Properties → Measurement → Data → Decision

Material

What material or materials are involved?

Students should learn to identify the substance, mixture, product, waste, or other hazardous material involved in a problem and determine what information is needed to characterize it.

Phase and Form

What physical form is the material in?

Relevant forms may include:

- Solid
- Liquid
- Gas
- Vapor
- Dust
- Fume
- Mist
- Aerosol

Understanding physical form helps determine how a material may move, how workers or the environment may encounter it, and how it should be measured or controlled.

Medium

Where is the hazardous material?

Potential media or compartments may include:

- Air
- Water
- Soil
- Surfaces
- Products
- Containers
- Waste streams

A hazardous material may move between multiple media during its lifecycle.

Movement

How can the material move or change?

Students should consider processes such as:

- Evaporation
- Dispersion
- Deposition
- Dissolution

- Leakage
- Spillage
- Transfer between containers
- Release to the environment

Properties

Which physical or chemical properties explain the material's behavior?

Chemistry and quantitative concepts should be introduced when they help students explain or solve a hazardous-material management problem.

Relevant concepts may include:

- Density
- Specific gravity
- Concentration
- Mass and volume
- Volatility
- Solubility
- Vapor pressure
- Chemical compatibility
- Reactivity

Calculations should support decision-making rather than exist as isolated mathematical exercises.

Measurement

How do we know how much material is present?

Students should learn how hazardous materials and exposures are measured and how measurement methods influence the data available for interpretation.

Laboratory activities may use videos, photographs, diagrams, demonstrations, and other resources to show how measurements are collected when physical sampling equipment is not available.

Data

What do the measurements tell us?

Students should learn to use data as evidence when evaluating hazardous-material problems.

The course introduces R and RStudio early so that students gain experience with a reproducible analytical environment. After the introductory R experience, students may use R, Excel, or another appropriate analytical interface for graded laboratory work unless a particular assignment requires a specific method.

Students may:

- Import data
- Inspect datasets
- Clean and organize variables
- Calculate descriptive statistics
- Perform relevant calculations
- Create tables
- Visualize measurements
- Compare observations
- Identify patterns
- Evaluate evidence

The analytical tool should be selected because it is appropriate for the hazardous-material problem rather than because a particular software package must be practiced.

R is one professional tool available for hazardous-material analysis rather than the primary subject of the course.

Decision

What should be done?

Students integrate scientific evidence, measurement data, technical guidance, regulatory requirements, and professional judgment to develop a hazardous materials management decision.

Potential decisions may involve:

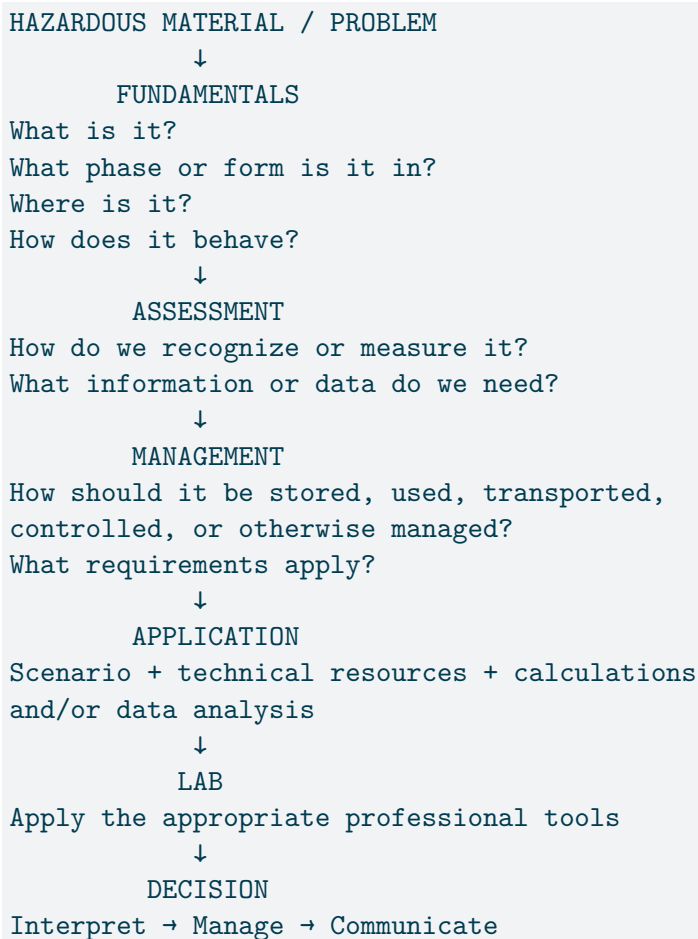
- Storage
- Handling
- Transportation
- Exposure controls
- Compatibility
- Inventory
- Spill prevention

- Emergency response
- Waste management
- Regulatory compliance

Module-Based Course Design

SHM 377 should be organized around integrated hazardous-material modules rather than treating scientific concepts, regulations, calculations, and laboratory activities as isolated course topics.

Each module should develop a hazardous-material problem through a recurring sequence:

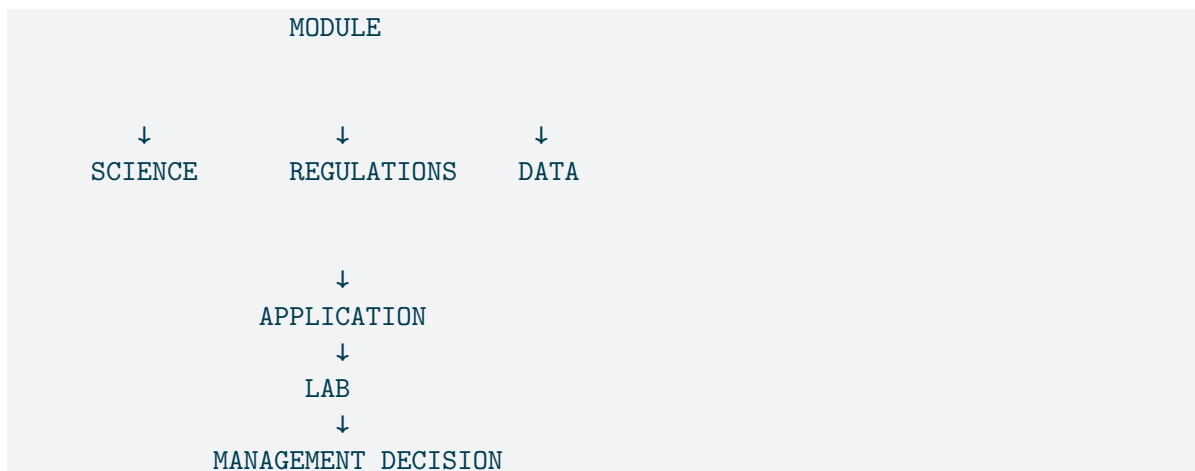


The amount of instructional time devoted to a module should depend on the complexity of the hazardous-material problem rather than requiring every module to occupy exactly one week.

Some modules may require only two class meetings, while more complex modules may span multiple weeks.

Integrated Module Components

Modules should integrate three major sources of evidence and reasoning:



The relative emphasis may change between modules.

A chemical-behavior module may emphasize physical and chemical properties and quantitative calculations.

A transportation module may emphasize classification and navigation of Department of Transportation resources.

An exposure or measurement module may emphasize sampling, measurement, and data analysis.

A storage module may emphasize compatibility, inventory, facility conditions, and applicable requirements.

An environmental-release module may emphasize movement between media, quantitative reasoning, reporting requirements, and environmental data.

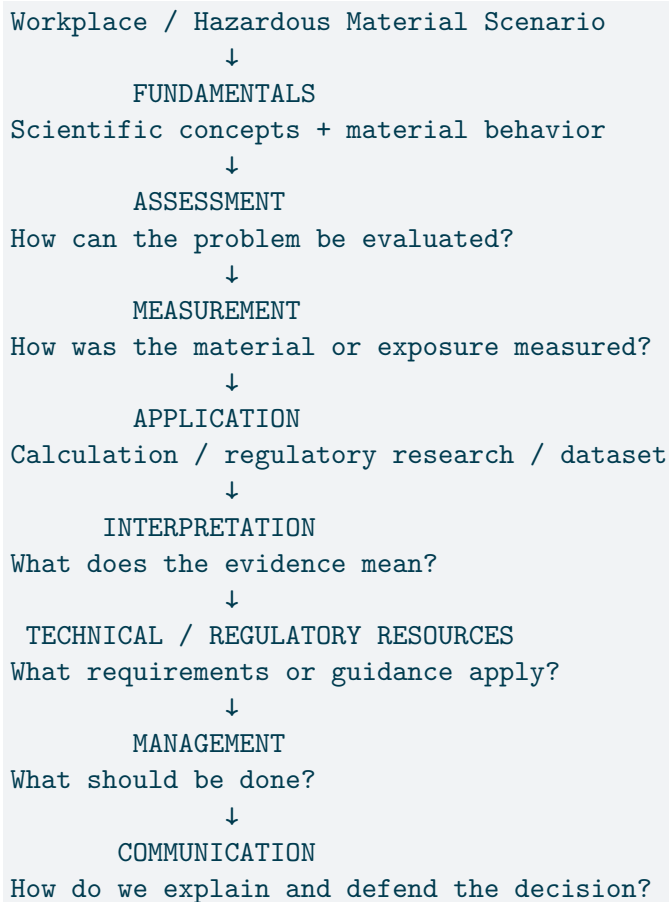
An emergency-response module may emphasize rapidly integrating incomplete information to support a defensible decision.

Students should nevertheless encounter the same underlying hazardous-material reasoning process repeatedly across these different problems.

Lecture–Laboratory Integration

SHM 377 combines lecture and laboratory activities into a common problem-solving sequence.

A typical instructional sequence may follow:



Lectures and laboratories should therefore reinforce one another rather than operate as separate components of the course.

Role of Chemistry

Chemistry provides tools for understanding hazardous-material behavior.

SHM 377 should build upon students' prerequisite chemistry knowledge without becoming a second general chemistry course.

Chemical concepts should be introduced in the context of questions such as:

- What is this material?
- What form is it in?
- Where is it?
- How might it move?
- What property explains that behavior?
- What calculation would help us understand the problem?
- What does that information mean for management?

Quantitative work should remain at an applied level appropriate for hazardous materials management.

Potential calculations include:

- Density
- Specific gravity
- Mass-volume relationships
- Concentration
- Unit conversions
- ppm
- Molarity
- Chemical quantities in containers
- Regulatory or reporting quantities

Students should learn to determine which calculation is appropriate based on the hazardous-material problem rather than simply selecting an equation from a list.

Role of Regulations

Regulations are professional resources rather than lists of facts to memorize.

Students should practice locating, navigating, interpreting, and applying regulatory requirements to hazardous-material problems.

Relevant regulatory frameworks may include requirements administered by:

- Occupational Safety and Health Administration
- U.S. Department of Transportation
- U.S. Environmental Protection Agency
- Other applicable federal or state agencies

Students should increasingly become responsible for determining:

```
What agency has authority?
↓
What regulation or technical resource applies?
↓
Where is the relevant requirement?
↓
What does it require?
↓
Does the scenario comply?
↓
What action is needed?
```

Regulatory navigation should be integrated into hazardous-material problems when the regulation becomes necessary for making or evaluating a management decision.

Role of R and Data Analysis

R and RStudio provide one reproducible environment for working with hazardous-material and exposure data.

Students are not expected to become programmers, and SHM 377 is not an R-programming course.

R is introduced early so that students gain experience using a professional analytical tool to import, inspect, summarize, visualize, and interpret a real environmental dataset.

After the introductory R experience, students may use R, Excel, or another appropriate analytical interface for graded laboratory work unless a particular assignment requires a specific method.

The emphasis is not on which software students use. The emphasis is on whether students can select an appropriate analytical approach, perform the analysis correctly, interpret the results, and use the evidence to support a defensible hazardous-materials management decision.

Students who choose to continue using R should be supported in doing so. Students who use another appropriate analytical tool should be held to the same standards for technical accuracy, documentation, interpretation, and communication.

The guiding principle is:

Support the mechanics. Assess the reasoning.

Role of Artificial Intelligence

Artificial intelligence is not prohibited in SHM 377. Students are encouraged to use AI thoughtfully as a tool for learning, problem solving, coding assistance, brainstorming, and improving their understanding.

Students remain responsible for the accuracy, interpretation, reasoning, and recommendations contained in work they submit. If AI is used in preparing a laboratory report, project, analysis, or other submitted work, students should briefly identify where AI was used and explain how or why it was used.

Students should not enter confidential, proprietary, sensitive, employer-owned, or otherwise restricted information into AI systems. AI availability and acceptable use also vary among workplaces. Students should therefore develop sufficient understanding of the underlying analytical methods and course concepts to evaluate AI-generated information and work without AI when necessary.

Instructional Philosophy

AI should support student thinking rather than replace it.

The purpose of SHM 377 is not to require students to memorize software syntax or complete technical work without access to modern tools. Students may receive R commands, Excel guidance, templates, examples, troubleshooting support, and other scaffolding throughout the course.

The emphasis is on whether students can:

- Understand the hazardous materials problem
- Understand what the data represent and how they were obtained
- Select or appropriately use analytical methods
- Interpret analytical results
- Evaluate the accuracy and limitations of information
- Apply technical and regulatory resources
- Develop defensible management recommendations
- Communicate their reasoning clearly

A useful principle for the course is:

Support the mechanics. Assess the reasoning.

Students should be able to explain the work they submit, including analyses or code developed with AI assistance.

Professional Use of AI

AI use in professional settings may be limited by organizational policies, information security requirements, confidentiality requirements, software restrictions, or restrictions on moving organizational data to personal devices or external systems.

SHM 377 should therefore help students learn both how to use AI appropriately and how to work when AI tools are unavailable or prohibited.

Students should develop enough understanding of data analysis, hazardous materials principles, and professional resources to evaluate AI-generated information rather than assuming that an AI-generated response is correct.

The final student-facing AI policy should be reviewed against current Central Washington University requirements before being incorporated into the course syllabus and other official course materials.

Initial R Workflow

Students should learn R commands in the context of answering questions about a dataset rather than learning programming syntax in isolation.

Virtual Lab 01: Introduction to R & Environmental Data serves as the primary introductory R experience and is also **Graded Lab 01**.

Virtual Lab 01 is worth **25 points** and is intentionally highly guided and completion based. Students who complete and submit the required materials receive full credit.

The purpose of Lab 01 is to establish the analytical and reproducible workflow used throughout the course rather than evaluate students' prior programming experience.

Students should complete Lab 01 having successfully:

- worked in RStudio;
- organized project files;
- created and saved an R script;
- used comments to organize an analysis;
- imported environmental monitoring data;
- inspected and checked a dataset;
- performed basic transformations and descriptive analyses;
- created a visualization;
- interpreted results cautiously;
- distinguished measurement from unsupported source attribution; and
- preserved a reproducible record of their analytical workflow.

Virtual Lab 01 also serves as a reference students can return to throughout the quarter when they need support with foundational R, RStudio, file-management, data-import, inspection, visualization, interpretation, or reproducibility skills.

Analytical and Professional Skill Progression

The four graded laboratories progressively reduce instructional scaffolding while increasing student responsibility for hazardous-material reasoning, analysis, interpretation, and management decisions.

```
LAB 01
Introduction to R & Environmental Data
Highly Guided / Completion Based
Establish the Analytical Workflow
    ↓
LAB 02
Guided Analysis
Apply the Established Workflow to a
Hazardous-Material Problem
    ↓
LAB 03
Semi-Structured Analysis
Make Analytical and Interpretive Choices
    ↓
LAB 04
Scenario-Based Decision Making
Select and Defend an Appropriate Approach
    ↓
FINAL GROUP PROJECT
Independent Team Integration
```

The progression reflects increasing independence in hazardous-material reasoning rather than increasing programming sophistication.

Lab 01 establishes the foundational analytical workflow. Later laboratories should build upon rather than reteach the basic R, RStudio, file-management, scripting, data-import, inspection, visualization, interpretation, and reproducibility skills introduced in Lab 01.

Students may use R, Excel, calculations, regulatory resources, technical references, or other appropriate tools depending on the problem and assignment requirements.

Laboratory Philosophy

SHM 377 laboratories should approximate the reasoning involved in real hazardous-material investigations even when physical sampling equipment is not available.

Where appropriate, laboratory activities should allow students to:

1. Observe a workplace, process, or hazardous-material problem.
2. Determine what information is needed.
3. Understand how measurements or samples were collected.
4. Work with relevant scientific, technical, regulatory, or quantitative information.
5. Analyze available evidence.
6. Interpret the results.
7. Develop a management recommendation.
8. Communicate the analysis in a professional format.

Datasets may include authentic public or open datasets when appropriate.

Professor-created or simulated datasets may also be used when they provide a clear instructional advantage. Simulated data should be clearly identified as such.

Virtual Laboratory Design Model

The finalized **Virtual Lab 01: Introduction to R & Environmental Data** serves as the instructional and visual model for subsequent SHM 377 virtual laboratories.

Where appropriate, virtual laboratories should follow a recognizable progression:



Hazardous-Material Significance



Management Decision



Reproducibility / Communication

The exact components and relative emphasis should vary according to the hazardous-material problem.

Lab 01 provides unusually detailed explanation because it establishes the technical workflow for students who may have little or no prior experience with R or RStudio.

Subsequent laboratories should preserve the same general instructional architecture while progressively reducing technical scaffolding and increasing the amount of hazardous-material reasoning expected from students.

Basic R concepts already established in Lab 01 should not normally be retaught from the beginning in later laboratories. Students should instead be directed back to Lab 01 as a reference when they need assistance with foundational analytical procedures.

The goal is for the analytical tools to become progressively less prominent while the hazardous-material problem, evidence, interpretation, regulatory context, and management decision become progressively more prominent.

Four Major Graded Laboratories

SHM 377 will include **four major graded laboratories** during the quarter.

Each graded laboratory is worth **25 points**, for a total of **100 laboratory points**.

The four laboratories collectively account for **50% of the total course grade**.

Lab 1	25 points
Lab 2	25 points
Lab 3	25 points
Lab 4	25 points
Total	100 points

The four graded laboratories should be intentionally selected based on the strength of the available hazardous-material problem, dataset, analytical opportunity, and professional application.

The course should not require a formal graded laboratory report every week.

Instead, the four major laboratories should provide students with sufficient time to investigate the problem, work with the relevant data or technical resources, practice the analytical methods, troubleshoot their work, interpret their findings, and produce a thoughtful final report.

Selecting the Four Laboratories

Potential graded laboratories should be evaluated based on:

1. Importance to SHM 377 learning outcomes.
2. Quality of the available dataset or technical problem.
3. Opportunity for meaningful quantitative or data analysis where appropriate.
4. Opportunity to integrate scientific and regulatory reasoning.
5. Professional relevance.
6. Ability to support a defensible management decision.
7. Contribution to skills needed for the final group project.
8. Appropriate level of challenge for an upper-division course.

The four laboratory topics should therefore be selected after candidate datasets, cases, and course modules have been evaluated.

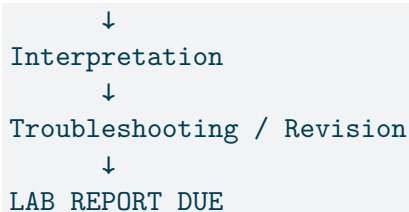
A topic should not become a graded laboratory merely because it occurs during a Wednesday meeting.

Laboratory Development Time

Students should receive meaningful time between the primary laboratory activity and the final report deadline.

A typical graded laboratory may follow:

```
MODULE / TOPIC
  ↓
Fundamentals
  ↓
Measurement / Assessment
  ↓
Guided Practice
  ↓
LAB INTRODUCED
  ↓
Explore Dataset / Problem in Class
  ↓
Appropriate Analytical / Technical Tools
```



Where practical, laboratory reports may be due approximately one week after the primary laboratory activity.

The exact due date should depend on the complexity of the assignment and the course calendar.

The purpose of this structure is to give students enough time to work with the material and analytical tools rather than requiring them to rush through a laboratory report immediately after encountering a new method or dataset.

Formative Laboratory and Class Activities

The four graded laboratories do not limit the amount of applied work completed during the course.

Other class meetings may include:

- R practice
- Excel or other data-analysis practice
- Dataset exploration
- Applied calculations
- Safety Data Sheet interpretation
- Regulatory navigation
- Hazard classification
- Storage and compatibility exercises
- Transportation decisions
- Environmental release scenarios
- Emergency-response scenarios
- Waste-characterization exercises
- Case studies
- Final-project development

These activities may be used for practice, feedback, preparation, and discussion without becoming additional major graded assignments.

This allows students to practice frequently while keeping the formal grading structure focused on a small number of substantial demonstrations of learning.

Laboratory Report Expectations

SHM 377 laboratory reports should be independently produced unless an assignment explicitly states otherwise.

Students may discuss concepts, methods, and troubleshooting with classmates, but each submitted report should reflect the student's own analysis, interpretation, calculations, and written work.

The four major graded laboratories should normally culminate in a professional written report.

A common report template should be introduced with Graded Lab 1 so that students can focus progressively more attention on analysis, interpretation, and decision-making rather than learning a new submission format for each assignment.

The standard report may include sections such as:

1. Scenario / Problem
2. Hazardous Material or Process
3. Available Evidence
4. Analysis
5. Results
6. Interpretation
7. Technical and Regulatory Considerations
8. Management Recommendations
9. Limitations
10. References

The exact sections may be modified when appropriate for a particular hazardous-material problem.

Students should normally submit the final report as a Word document unless an assignment specifies another format.

Analytical work may be completed using R, Excel, calculations, regulatory resources, or another appropriate method.

Laboratory reports should meet the following expectations:

- Written responses should use complete sentences where interpretation or explanation is required.
- Tables and figures should be clearly labeled.
- Graphs should include informative axis labels and appropriate units.
- Units should be included in calculations and final answers.
- Calculations should be shown or documented clearly enough that the reasoning can be followed.

- Numerical results should be interpreted in the context of the hazardous-material problem rather than reported without explanation.
- Data sources, technical references, and regulatory resources should be cited when appropriate.
- Reports should distinguish results from interpretation and management recommendations.
- Reports should identify the student and laboratory.
- Files should be submitted through Canvas using the required naming convention.

Students should not submit unexplained computer output in place of written interpretation.

Documentation of Analytical Work

Regardless of the analytical tool used, students should document their work sufficiently for the professor to understand how the results were obtained and evaluate whether the analysis and reasoning are technically defensible.

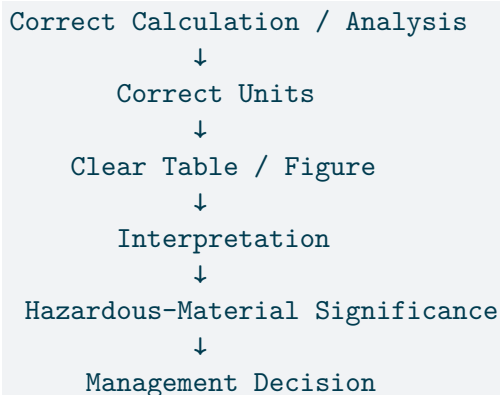
Where R is used, students should preserve enough of the analytical workflow to make the results traceable and, where appropriate, reproducible.

Where Excel is used, students should organize calculations, formulas, tables, and figures clearly enough for the analytical process to be followed.

If a calculation is completed manually, students should show the appropriate setup and units.

If another analytical tool is used, the student remains responsible for providing enough documentation to explain the method and support the reported results.

A laboratory report should progress beyond producing a correct number or graph:



Laboratory Methods

Not every SHM 377 laboratory needs to use the same analytical method.

The laboratory method should be selected according to the hazardous-material problem and the professional task students need to practice.

Laboratory activities may involve:

- R and quantitative data analysis
- Excel or other appropriate data-analysis tools
- Measurement-data interpretation
- Applied chemistry calculations
- Safety Data Sheet interpretation
- Regulatory navigation
- Hazard classification
- Storage and compatibility evaluation
- Transportation and shipping decisions
- Facility or inventory assessment
- Environmental release analysis
- Emergency-response scenarios
- Waste-characterization problems
- Technical communication
- Combinations of these approaches

The purpose of the laboratory is not to practice a particular tool for its own sake.

The purpose is to select and use appropriate tools to investigate and manage a hazardous-material problem.

Provisional Module Architecture

The following module structure is a planning framework rather than a finalized course schedule.

Potential modules include:

Module	Scientific / Technical Focus	Potential Application
Course & Analytical Foundations	Hazardous-material reasoning, R/RStudio, data interpretation	Graded Lab 01: Introduction to R & Environmental Data
Chemical Behavior	Phase, form, medium, properties, calculations	Chemical case

Module	Scientific / Technical Focus	Potential Application
Measurement & Exposure	Sampling, measurement, exposure data	Monitoring dataset
Radiation	Fundamentals, units, behavior, protection	Radiation case/data
Storage & Use	Compatibility, inventory, controls	Facility scenario
Transportation	Classification, DOT resources, shipping	Transportation case
Environmental Release	Release, movement, environmental data	Release analysis
Emergency Response	Spill/release assessment and response	Incident scenario
Hazardous Waste	Characterization, management, disposal	Waste case
Integration	Multi-hazard decision-making	Final group project

This sequence is intentionally provisional.

The final module order should be evaluated against:

1. SHM 377 catalog learning outcomes.
2. Required hazardous-material management content.
3. The strengths and limitations of inherited SHM 377 materials.
4. Availability of appropriate cases, datasets, and laboratory activities.
5. Logical progression of scientific and quantitative concepts.
6. Regulatory progression.
7. Student workload.
8. Final-project skill requirements.

Topics do not need to receive equal amounts of instructional time.

More complex modules may span multiple weeks, while narrower topics may require fewer class meetings.

Fall 2026 Course Architecture

The Fall 2026 course schedule is organized around the Central Washington University academic calendar and the Monday–Wednesday meeting pattern.

Course modules, four major graded laboratories, formative activities, and final-project milestones will be mapped onto this calendar as the course is developed.

The detailed topic sequence remains provisional.

Week 1 — September 23

Because Fall Quarter instruction begins on Wednesday, SHM 377 begins with a single scheduled meeting during the first partial week.

Lecture 01: Introduction to Hazardous Materials Management

Lecture 01 establishes the conceptual foundation for the course and introduces students to hazardous materials management as a systems problem.

Lecture 01 is complete and locked for Fall 2026.

Week 2 — September 28 & 30

Initial analytical foundation:

- Virtual Lab 01: Introduction to R & Environmental Data
- R and RStudio orientation
- Course technology and analytical workflow
- Importing, inspecting, summarizing, and visualizing environmental data
- Interpreting measurements in the context of a hazardous-material problem
- Distinguishing measurement evidence from unsupported source attribution

Virtual Lab 01 is **Graded Lab 01** and is worth **25 points**.

The laboratory is highly scaffolded and completion based. Its purpose is to ensure that students successfully establish the course analytical workflow and have an accessible first laboratory grade before later assignments require greater independence.

Students who complete and submit the required Lab 01 materials receive full credit.

After Virtual Lab 01, course development should shift toward substantive hazardous-material lectures and modules. R may continue to be used where appropriate, but subsequent graded laboratories should not be framed as an R-coding sequence.

Week 3 — October 5 & 7

Under development

Week 4 — October 12 & 14

Under development

Week 5 — October 19 & 21

Under development

Week 6 — October 26 & 28

Under development

Week 7 — November 2 & 4

Under development

Week 8 — November 9

Veterans Day: Wednesday, November 11 — No class

Only the Monday meeting is scheduled during this week.

Week 9 — November 16 & 18

Under development

Week 10 — November 23

Thanksgiving Recess: November 25–27 — No class

Only the Monday meeting is scheduled during this week.

Week 11 — November 30 & December 2

Final-project presentations and course synthesis — provisional

End-of-Quarter Period — December 4–11

December 4: Last day of instruction

December 7: Study Day

December 8–11: Final examination period

Final examination-period activities will be determined as the final-project schedule is developed.

Assessment Architecture

SHM 377 uses a **200-point course grading system**.

The course grade is divided equally between four major applied laboratories and the final group project.

Assessment	Points	Percent of Course Grade
Graded Lab 1	25	12.5%
Graded Lab 2	25	12.5%
Graded Lab 3	25	12.5%
Graded Lab 4	25	12.5%
Laboratory Total	100	50%
Final Group Project	100	50%
Course Total	200	100%

The assessment architecture is intentionally simple:

```
FOUR MAJOR LABS
4 × 25 points
  ↓
100 points
50% of course grade

      +

FINAL GROUP PROJECT
100 points
50% of course grade

  ↓
```

TOTAL
200 points

The four laboratories provide structured opportunities for students to practice and demonstrate the scientific, quantitative, regulatory, analytical, and professional reasoning developed throughout the course.

The final group project asks students to integrate those capabilities in a more independent hazardous-material investigation.

Assessment Progression

The graded assessments should progressively move from structured to independent hazardous-materials problem solving:

```
Lab 01  
Highly Guided / Completion Based  
  ↓  
Lab 02  
Guided  
  ↓  
Lab 03  
Semi-Structured  
  ↓  
Lab 04  
Increasing Independence  
  ↓  
Final Group Project  
Independent Team Integration
```

The four graded laboratories should therefore function as preparation for the final group project rather than as four unrelated assignments.

The progression should reflect increasing independence in identifying needed information, selecting appropriate tools, analyzing evidence, applying technical and regulatory resources, interpreting results, and developing management recommendations.

Formative Work

Regular class activities do not need to carry course points in order to contribute meaningfully to learning.

Students may regularly complete:

- Practice calculations
- R exercises
- Excel or other data-analysis exercises
- Regulatory searches
- Data interpretation
- Case-study questions
- Group problem solving
- Technical-resource navigation
- Draft analyses
- Final-project development

These activities should provide opportunities to practice, receive feedback, ask questions, and prepare for the four graded laboratories and final group project.

The course should avoid creating unnecessary graded work merely to generate points.

Final Group Project

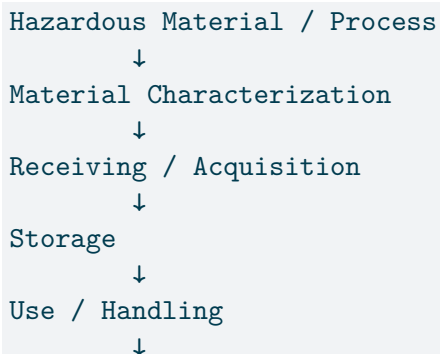
The final group project is worth **100 points** and accounts for **50% of the total course grade**.

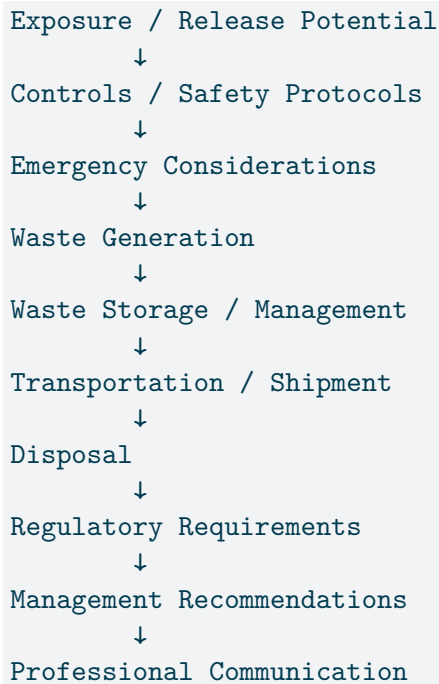
The project serves as the culminating application of the SHM 377 hazardous materials reasoning framework.

Students will work in teams to investigate a complex hazardous-materials management scenario provided by the professor.

The scenario should require students to integrate multiple stages of hazardous materials management rather than solve a single isolated analytical problem.

A project may require teams to consider:





Students should determine what information is needed, locate appropriate technical and regulatory resources, evaluate available evidence, perform appropriate analyses where necessary, identify limitations or missing information, and develop defensible management recommendations.

The project should assess integration and professional reasoning rather than require the use of a particular analytical software package.

Potential deliverables may include:

- Professional written report
- Figures and tables where appropriate
- Applied calculations where appropriate
- Data analysis where appropriate
- Regulatory analysis
- Management recommendations
- Oral presentation

The exact internal distribution of the 100 final-project points will be developed with the final-project rubric.

The final-project scenario, grading rubric, team expectations, and development milestones should be established before students begin substantial project work.

A potential scenario model is an organization preparing to receive and use a hazardous material that requires students to consider receiving, storage, use, worker protection, emergency planning, waste management, transportation, and disposal across the material lifecycle.

Final Group Project Skill Progression

The four graded laboratories and other course modules should progressively develop capabilities students will eventually need for the final group project.

A general progression may follow:



Final-project development should therefore be integrated with course modules rather than treated as an unrelated end-of-quarter assignment.

Case Study Bank

Potential case studies should allow students to integrate multiple components of hazardous-material management rather than solve isolated problems.

Cases may involve:

- Workplace chemical use
- Storage and compatibility
- Transportation
- Worker exposure
- Environmental releases
- Emergency response
- Hazardous waste
- Radiological materials
- Biological materials
- Multi-hazard facilities
- Aerospace or advanced-manufacturing scenarios involving regulated materials

Where possible, cases should include enough information for students to practice the full reasoning framework.

Cases should vary in the tools required to solve them.

A potential integrative final group project scenario could place students in an aerospace or advanced-manufacturing organization preparing to receive and use uranium or another tightly regulated hazardous material.

Students could be asked to determine what is needed for receiving, storage, use, worker protection, emergency planning, waste management, transportation, and disposal.

This remains a scenario model rather than a locked Fall 2026 project topic.

Some cases may primarily require quantitative analysis.

Others may primarily require regulatory research, material classification, compatibility assessment, technical interpretation, or professional judgment.

Dataset Bank

Datasets should support authentic or realistic hazardous-material management questions.

Potential dataset categories include:

- Air monitoring
- Surface contamination
- Environmental releases
- Chemical inventory
- Storage
- Waste generation
- Transportation
- Incident or spill data

- Exposure measurements
- Environmental monitoring

Each dataset should document:

- Source
- Whether the data are authentic or simulated
- Variable definitions
- Units
- Measurement context
- Intended learning objectives
- Applicable limitations

Datasets should be selected because they support a meaningful hazardous-material question rather than simply because they provide an opportunity to use R.

The quality and instructional usefulness of candidate datasets should be one of the primary considerations when selecting the four graded laboratory topics.

Student Onboarding

Welcome to SHM 377

A student-facing orientation should introduce students to:

- Course purpose
- Monday–Wednesday course structure
- Module-based course organization
- Lecture and laboratory integration
- Hazardous-material reasoning framework
- Four-laboratory assessment structure
- Final group project
- Course repository
- R and RStudio
- Excel or other appropriate analytical tools
- Expectations for quantitative work
- Expectations for regulatory research

Students should understand that the course contains only four major graded laboratories, but applied work and laboratory-style activities will occur throughout the quarter.

The first weeks should also establish that students are not expected to enter the course already knowing R.

Students should understand that R is one of several professional tools they will use throughout the course.

After the introductory R laboratory, students should understand that they may use R, Excel, or another appropriate analytical interface for graded work unless a specific assignment requires a particular method.

Introductory R and RStudio Setup

Students should receive clear instructions for:

1. Installing or accessing R.
2. Installing RStudio.
3. Opening course files.
4. Running provided code.
5. Accessing course datasets.
6. Inspecting a dataset.
7. Performing basic descriptive analysis.
8. Creating a basic visualization.
9. Interpreting and documenting the analytical result.

Early activities should emphasize successful use of the tools rather than programming syntax.

Interactive Learning

Interactive activities should be incorporated where they advance the learning objectives.

Potential activities include:

- Case-study questions
- Material identification
- Phase/form classification
- Selecting appropriate measurements
- Interpreting Safety Data Sheets
- Regulatory navigation
- Calculation problems
- Data interpretation
- Small-group decision-making
- Management recommendations
- Facility scenarios
- Transportation decisions
- Emergency-response decisions

Activities should increasingly move from recognition toward application and decision-making.

A useful pattern is:

```
graph TD; Observe --> Ask; Ask --> Investigate; Investigate --> Analyze; Analyze --> Interpret; Interpret --> Decide; Decide --> Defend;
```

Observe
↓
Ask
↓
Investigate
↓
Analyze
↓
Interpret
↓
Decide
↓
Defend

Standard Module Components

To maintain consistency across SHM 377 while allowing different topics to receive different amounts of instructional time, modules should generally consider the following components.

Hazardous-Material Problem

What problem, material, facility, process, or decision anchors the module?

Fundamentals

What scientific or technical concepts are needed to understand the problem?

Assessment

What information would a professional need to evaluate the problem?

Measurement and Evidence

How can relevant information be measured, observed, calculated, or otherwise obtained?

Technical and Regulatory Resources

What professional resources are needed?

Application

What should students do with the information they have gathered?

Laboratory

What professional task or tool should students practice?

Does this module provide a sufficiently strong problem, dataset, and professional application to justify being selected as one of the four major graded laboratories?

Management Decision

What should be done and why?

Communication

How should the analysis and recommendation be documented or defended?

Standard Lecture Components

To maintain consistency across SHM 377, lectures should use a common set of structural elements where appropriate.

Course Identification

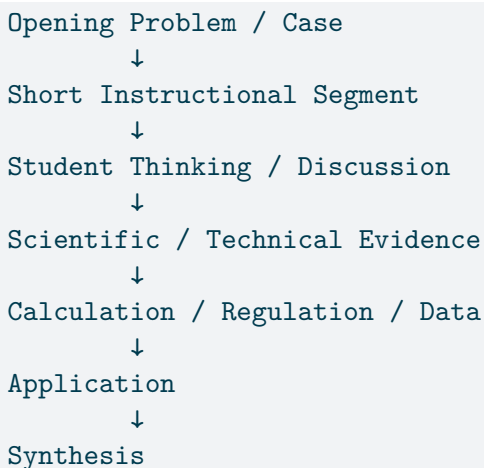
Lecture materials should identify:

- **SHM 377: Hazardous Materials Management**
- Occupational Safety and Health Management
- Yoni Rodriguez
- Engineering Technologies, Safety, and Construction
- Central Washington University
- Yoni.Rodriguez@cwu.edu

Instructional Structure

Because SHM 377 meets in extended instructional periods, meetings should not assume continuous professor presentation.

A meeting may combine multiple instructional segments:



The exact structure should remain flexible based on the instructional purpose of the meeting and the module.

Hazardous Materials Reasoning

Where appropriate, lectures should return to the common reasoning framework:

Material → Phase → Medium → Movement → Properties → Measurement → Data → Decision

Students should progressively become responsible for applying more of this framework themselves as the quarter progresses.

Lecture–Laboratory Connection

Lectures should clearly identify connections to laboratory activities where appropriate.

Students should understand why a scientific concept, calculation, regulation, measurement method, or technical resource introduced during lecture is needed for the associated application or laboratory problem.

Lecture Closing

Where appropriate, lectures should conclude by reinforcing major connections among:

- Material behavior
- Measurement
- Evidence
- Data
- Regulations
- Management decisions

A closing may:

- Return to the opening hazardous-material problem
- Revisit the reasoning framework
- Ask students what information is still missing
- Ask students what regulation or technical resource they would need
- Ask students what management decision they would make
- Preview the laboratory or next stage of the module
- Identify key vocabulary
- Provide a synthesis question

Lecture materials should also contain the standard copyright and use statement.

Standard Laboratory Components

Where appropriate, SHM 377 laboratories should contain the following components.

Problem or Scenario

What hazardous-material problem are we investigating?

Purpose

What professional question or decision is the laboratory intended to address?

Measurement or Information Context

How were the data generated, measurements collected, or technical information obtained?

Evidence

What information is available?

This may include:

- Dataset
- Measurements
- Safety Data Sheet
- Facility information
- Inventory
- Shipping information
- Regulatory text
- Photographs
- Process diagrams
- Incident information
- Technical guidance

Analysis

What calculations, comparisons, classifications, summaries, visualizations, or other analyses are needed?

Interpretation

What does the evidence mean?

Technical and Regulatory Context

What additional resources or requirements should be considered?

Management Decision

What action would you recommend and why?

Communication

How should the findings and recommendation be documented?

Course Development Quality Checks

Before a lecture, laboratory, case study, module, or assessment is considered complete, it should be reviewed through several lenses.

Student Accessibility

Can a student with the expected prerequisite background understand what is being asked?

Technical Depth

Is the material appropriate for an upper-division SHM course?

Professional Relevance

Does the activity resemble a question or decision a safety or hazardous materials professional could reasonably encounter?

Regulatory Currency

Are regulatory requirements, guidance documents, tables, and references current?

Quantitative Solvability

Do students have the information, units, equations, and background needed to complete required calculations?

Lecture–Laboratory Alignment

Does the laboratory apply or extend concepts students have encountered during the associated lecture or module?

Tool Appropriateness

Is the laboratory using R, Excel, calculations, regulatory research, or another tool because it is appropriate for the problem rather than simply because the tool needs to be practiced?

Workload

Is the expected student workload appropriate for the course and available laboratory time?

For the four graded laboratories, does the due date provide sufficient time for students to explore the problem, troubleshoot their analysis, interpret their findings, and prepare a professional report?

Learning-Outcome Alignment

Does the activity contribute meaningfully to one or more SHM 377 learning outcomes?

Module Coherence

Do the scientific concepts, regulations, data, laboratory activity, and management decision contribute to the same hazardous-material problem?

Final-Project Alignment

Does the activity develop a capability students will eventually need for the final group project?

Graded Laboratory Value

If the activity is being considered as one of the four graded laboratories, does it provide enough analytical depth, professional relevance, and integration to justify 25 points and substantial student work?

Open Design Questions

The following questions remain intentionally unresolved during the continuing course-development phase:

- What is the final module sequence?
- Which modules require multiple weeks?
- Which three additional modules provide the strongest problems, scenarios, datasets, and professional applications for Graded Labs 02–04?
- What should the due date be for each graded laboratory?
- Which graded laboratories should emphasize data analysis, regulatory navigation, calculations, classification, facility assessment, or other professional methods?

- Which datasets should anchor the graded laboratories where quantitative data are appropriate?
- Where should radiological and biological hazards appear in the sequence?
- How should Hazard Communication and GHS be integrated across modules?
- How should federal hazardous-material regulations be introduced and revisited?
- What complex professor-provided scenario should anchor the Fall 2026 final group project?
- How should the 100 final group project points be distributed across the project deliverables?
- How should individual accountability be incorporated within the group final project?
- How should peer review contribute to final group project development?
- Which inherited SHM 377 materials should be retained, modified, or replaced?
- Which inherited regulatory resources need to be updated before Fall 2026?

The following course-design decisions are now established:

- The course uses a **200-point grading system**.
- There are **four major graded laboratories**.
- Each graded laboratory is worth **25 points**.
- The four laboratories total **100 points** and account for **50% of the course grade**.
- The final group project is worth **100 points** and accounts for **50% of the course grade**.
- Weekly formal laboratory reports are not required.
- Other laboratory and class activities may be formative rather than graded.
- **Virtual Lab 01: Introduction to R & Environmental Data is Graded Lab 01.**
- Graded Lab 01 is worth **25 points**.
- Graded Lab 01 is intentionally **highly guided and completion based**.
- Students who complete and submit the required Lab 01 materials receive full credit.
- Lab 01 is designed to establish a common analytical and reproducible workflow rather than assess prior programming experience.
- The finalized Virtual Lab 01 serves as the instructional and visual model for subsequent SHM 377 virtual laboratories.
- Lab 01 serves as a reference for foundational R, RStudio, file-management, scripting, data-import, inspection, visualization, interpretation, and reproducibility skills.
- Subsequent laboratories should build upon rather than reteach the foundational analytical skills established in Lab 01.
- Laboratory scaffolding should progressively decrease from Lab 01 through Lab 04 while expectations for hazardous-material reasoning, interpretation, and management decisions progressively increase.
- SHM 377 is not an R-programming course.
- After the introductory R experience, students may use R, Excel, or another appropriate analytical interface unless an assignment requires a particular method.
- Analytical tools should be selected according to the hazardous-material problem rather than required for their own sake.

- The four graded laboratories should progressively develop hazardous-materials reasoning and professional independence rather than R-programming independence.
- Graded Lab 1 should establish a common professional report template that can be carried forward through later laboratories.
- Major graded laboratory reports should normally be submitted as professional written reports, ordinarily as Word documents unless otherwise specified.
- The four graded laboratories should be selected based on instructional value, dataset or evidence quality, professional relevance, and alignment with the final group project.
- Students should receive meaningful development time for major laboratory reports.
- Laboratory scaffolding should progressively decrease as students move toward the final group project.
- The final project will be completed as a **group project**.
- The final group project will use a complex **professor-provided hazardous materials management scenario**.
- The final group project should require integration across multiple stages of the hazardous-material lifecycle.
- The four graded laboratories should prepare students progressively for the reasoning required in the final group project.
- Future course development should prioritize substantive lecture and module development while laboratory scenarios are selected as strong applications emerge.

Copyright, Licensing, and Attribution

Course Materials

Original SHM 377 materials created for the course should clearly identify authorship and copyright.

Except where otherwise noted, original educational materials are licensed under the **Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)**.

Third-party materials incorporated into lectures and laboratories remain subject to their respective copyright, licensing, attribution, and permitted-use requirements.

The presence of third-party material within an SHM 377 course resource does not place that material under the license applied to original SHM 377 content.

Standard Lecture and Laboratory Footer

Each lecture and laboratory should conclude with a copyright and use statement similar to:

Copyright & Use

****SHM 377: Hazardous Materials Management****

© 2026 Yoni Rodriguez

Except where otherwise noted, original course materials are licensed under the Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0).

Third-party materials remain subject to their respective copyright and licensing terms.

Attribution

Sources used in course materials should be appropriately attributed.

Attribution is particularly important for:

- Figures
- Photographs
- Maps
- Tables
- Government resources
- Regulations and regulatory guidance
- Published research
- Technical documents
- Safety Data Sheets
- Case-study materials
- External datasets
- Adapted diagrams or frameworks

Where practical, original course materials should distinguish between:

- Original professor-created content
- Adapted content
- Third-party content reproduced under an applicable license or permitted use

Course Repository

The SHM 377 course repository is intended to provide students with access to course materials and reproducible analytical resources.

The repository includes a `LICENSE.md` file documenting copyright, licensing, attribution, and reuse terms.

Except where otherwise noted, original course materials are licensed under the **Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)**.

The course license does not automatically apply to third-party materials contained within or referenced by the repository. Third-party materials remain subject to their respective copyright and licensing terms.

Public Repository Materials

The public repository may include:

- Lecture `.qmd` files
- Rendered lecture HTML files
- Laboratory `.qmd` files
- Rendered laboratory HTML files
- Student-facing analytical code and templates
- Publicly distributable datasets
- Professor-created figures and graphics
- Professor-created photographs
- Student-facing assignment instructions
- Final-project instructions and rubrics
- Course vocabulary and study materials
- Repository documentation
- `LICENSE.md`

Restricted Materials

The public repository should not include:

- Student information
- Student-generated data containing identifying information
- Grades
- Restricted examination questions
- Answer keys

- Previous student submissions
- Copyrighted third-party materials that cannot legally be redistributed
- Internal university records
- Protected course records

Restricted materials should be maintained separately from the public teaching repository.

Professor Notes and Reflections

Fall 2026

This section should be updated during and after the first offering of the redesigned SHM 377 course.

Keep

Elements that worked particularly well and should be retained in future offerings.

Modify

Elements that were useful but should be revised.

Remove

Elements that did not meaningfully contribute to student learning or course goals.

Ideas for Future Offerings

New modules, cases, datasets, laboratory activities, regulatory exercises, technologies, assessment approaches, or structural changes to consider in future versions of SHM 377.